



Review

Towards automated quality assurance of patent specifications: A multi-dimensional LLM framework[☆]Yujuan Chai^{*}, Chaochao Wang, Weilei Wang

Patsnap, Suzhou, China

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ABSTRACT

Despite the widespread adoption of Large Language Models (LLMs) in intellectual property domains, patent evaluation has remained unexplored and continues to rely on labor-intensive manual processes. To address this gap, we propose a multi-dimensional LLM framework for automated patent specification quality assurance. Our framework comprises three specialized evaluation modules: (1) a regulatory compliance module assessing adherence to patent office guidelines, (2) a technical coherence module evaluating logical consistency and technical soundness, and (3) a figure–reference consistency module verifying alignment between textual descriptions and drawings. A final integration step synthesizes module-level defect signals into prioritized, actionable revision recommendations, bridging automated assessment with practical patent drafting and pre-examination workflows. The framework is evaluated on 160 patent specifications, consisting of 80 human-authored CNIPA patents and 80 AI-generated specifications produced by commercial drafting tools. Three evaluation modules achieve balanced accuracies of 99.74%, 82.12%, and 91.2% respectively against expert annotations. Additional analysis examines defect distributions across patent sections, technical domains, and authoring sources, providing further insights into quality patterns of LLM-generated patent specifications.

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^{*} Corresponding author.

E-mail addresses: chaiyuqian@patsnap.com (Y. Chai), wangchaochao@patsnap.com (C. Wang), wangweilei@patsnap.com (W. Wang).

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1. Introduction

With over 3.55 million patent applications filed globally in 2023,¹ patent examination systems face growing scalability challenges in quality assessment. Although expert review remains the gold standard for evaluating patents, growing application volumes make purely manual examination costly and difficult to sustain. This has intensified interest in automated approaches to patent quality assessment. At the same time, the emergence of AI-powered patent drafting tools has further reshaped patent preparation practices [1]. While such tools significantly improve drafting efficiency, they also introduce new quality risks, including technical inaccuracies, insufficient disclosure, and inappropriate legal terminology [1,2]. If left unaddressed, these defects may undermine legal validity and weaken the patent system's fundamental role in promoting technological disclosure and knowledge dissemination [3,4].

Despite this urgent need, existing automated patent evaluation methods provide limited support for systematic, document-level quality assurance. Early approaches rely primarily on aggregate proxy indicators, such as citation patterns and legal status [5,6], while more recent AI-driven methods focus on patent value prediction using machine learning models [7,8]. Although some studies [9] incorporate document content, they typically depend on generic NLP metrics that do not capture patent-specific requirements. Together, these limitations highlight the absence of frameworks explicitly designed for holistic quality evaluation of complete patent specifications.

Large language models (LLMs) have demonstrated strong capabilities in patent-related tasks, including prior art search, summarization, and analysis [10–12]. Emerging work has begun to apply LLMs to patent quality assessment, primarily at the claim level [13]. However, systematic document-level evaluation of patent specifications remains largely unexplored. Recent advances in multimodal LLMs enable joint reasoning over textual and visual content [14], providing the technical foundation for unified assessment of regulatory compliance, technical coherence, and figure–text consistency within complete patent documents.

This paper presents an automated quality assurance framework for patent specifications. The main contributions are as follows:

- (1) **Three-dimensional assessment framework:** an LLM-driven evaluation system that jointly assesses regulatory compliance, technical coherence, and figure–text consistency at the document level;
- (2) **Actionable improvement integration:** an integration mechanism that consolidates module-level defect signals into prioritized and concrete revision actions to support patent drafting and pre-examination workflows;
- (3) **Empirical validation across domains:** a comprehensive evaluation on 160 patent specifications spanning eight technical domains, demonstrating robust performance across patent types and complexity levels;

(4) **Defect pattern analysis:** a systematic analysis of quality defect distributions across patent sections, technical domains, and authoring sources.

2. Related work

2.1. Automated patent assessment

Patent quality assessment has traditionally relied on human experts, resulting in high costs, long evaluation cycles, and subjective judgments [5]. Early automated approaches therefore focused on indirect proxy indicators, such as citation patterns, claim scope, and legal status, as signals of patent quality [5,6]. While later studies proposed indicator-based frameworks incorporating technical and legal attributes [7,15,16], these methods largely operate at a macro level and are not designed to assess document-intrinsic quality of patent specifications.

With the emergence of LLMs, patent assessment research has begun to adopt semantically informed evaluation strategies. Recent work increasingly follows the LLM-as-Judge paradigm to approximate expert judgments, most notably for patent claim evaluation [13,17]. For example, Yoo et al. [13] demonstrate that LLM-based claim scoring correlates more strongly with expert judgments than traditional NLP metrics.

However, existing LLM-based assessment efforts remain primarily formulated at the claim level, leaving document-level quality evaluation of complete patent specifications largely unexplored.

2.2. LLM applications in patent drafting

LLMs have been increasingly explored for patent drafting in both academic research and commercial systems [18]. Prior studies demonstrate that models fine-tuned on patent corpora can generate patent claims [19,20], abstracts [21], and even complete patent specifications [9]. In parallel, commercial platforms such as MindFlowing, PatSnap, and XLScout have incorporated LLM-based drafting functionalities into broader patent preparation workflows. Existing studies and systems tend to emphasize LLM-based patent generation capability and workflow efficiency. In contrast, quality validation is often treated in a relatively lightweight manner, relying on surface-level metrics, task-specific benchmarks, or small-scale expert assessments, which offer limited insight into document-level quality risks.

At the same time, prior work [22] has shown that LLM-generated text may exhibit hallucinations, factual inconsistencies, and subtle logical errors, which are particularly problematic in legal and technical documents such as patent specifications. As LLM-assisted patent drafting becomes more prevalent, these characteristics pose increasing challenges for patent examination and quality control, highlighting the need for systematic, document-level quality assurance frameworks to assess LLM-generated patent specifications in comparison with human-authored ones.

¹ https://www.wipo.int/pressroom/en/articles/2024/article_0015.html.

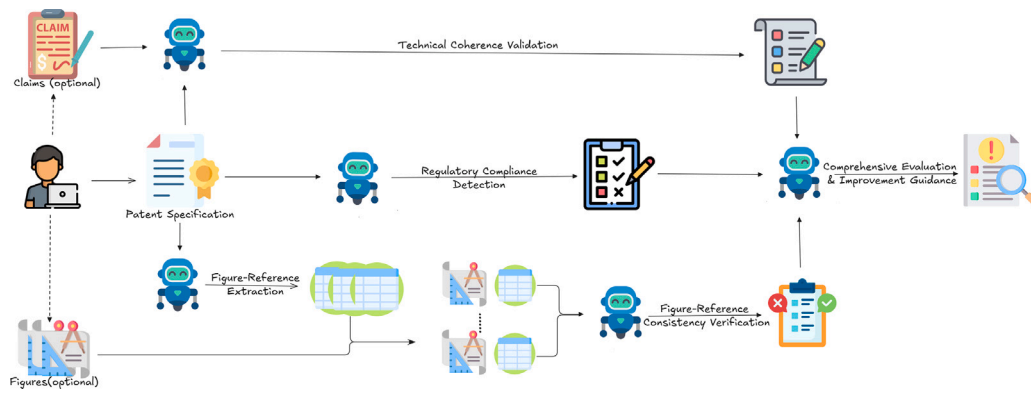


Fig. 1. Overview of the patent quality assurance framework. Three specialized detection modules operate in parallel to identify regulatory compliance, technical coherence, and figure–reference consistency issues, followed by an integration module that synthesizes findings into improvement recommendations.

3. Methodology

We adopt a state-of-the-art large language model as the evaluation engine for all modules.² As shown in Fig. 1, the framework consists of three parallel detection modules followed by an integration module.

3.1. Regulatory compliance detection

The regulatory compliance module formalizes drafting requirements defined by patent examination guidelines. In this study, rule instantiation follows CNIPA standards, while guideline-specific constraints can be modified to accommodate alternative jurisdictions.

Six categories of compliance violations frequently observed in specification review are modeled: (1) prohibited promotional or commercial language (e.g., subjective claims such as “revolutionary breakthrough”); (2) inconsistent technical terminology and naming conventions; (3) omission of mandatory structural components required by examination practice; (4) improper title or abstract formatting; (5) orthographic or typographical errors that may affect legal interpretation; and (6) insufficient figure descriptions that compromise disclosure completeness.

Each sentence is evaluated with a binary compliance decision. When non-compliant, the module provides a concise explanation referencing the relevant guideline principle.

3.2. Technical coherence validation

The technical coherence module evaluates internal consistency and enablement-related quality at the sentence level. It models defects commonly observed in specification review, including claim–specification contradictions, insufficient technical support, terminology inconsistencies, logical discontinuities affecting reproducibility, and conflicts with cited prior art. The module also accounts for fabrication patterns observed in AI-generated text, such as unsupported technical details and artificially constructed reasoning sequences.

Each sentence is classified into one of four risk levels: *Safe*, *Low*, *Medium*, or *High*. Risk levels reflect the potential impact of the defect on patentability and examination outcome. High-risk cases include fundamental technical contradictions, fabricated details, technical-field mismatches, missing essential claim elements, or figure-description inconsistencies. Medium-risk cases include insufficient claim support, incomplete technical features, imprecise scope definitions, or weak

Table 1

Risk classification criteria and examples for technical coherence validation.

Risk level	Examples
High	- Contradicts fundamental theories - Includes fabricated technical details - Mismatches technical field - Lacks technical details in claims - Lacks essential figures - Shows completely inconsistent drawing labels or descriptions - Constructs artificial reasoning chains
Medium	- Defines field scope imprecisely - Cites literature with low relevance - Provides insufficient support for claims - Lacks depth in technical descriptions - Establishes improper logical connections - Lacks proper experimental validation - Omits necessary views or includes irrelevant ones
Low	- Cites dated but valuable literature - Shows minor expression differences from claims - Contains typos not affecting understanding - Has minor formatting issues - Uses slightly imprecise terminology

logical connections. Low-risk cases include minor expression differences, limited terminology imprecision, or typographical errors that do not materially affect technical meaning. Representative classification criteria and examples are summarized in Table 1.

3.3. Figure–reference consistency verification

The figure–reference consistency module evaluates alignment between graphical elements and textual descriptions.

The module proceeds in three stages: (i) reference extraction to construct numeral inventories for each figure; (ii) bidirectional alignment to verify correspondence between textual references and graphical numerals; and (iii) cross-figure validation to distinguish genuine inconsistencies from legitimate structural variations.

A figure–reference pair is considered *consistent* when numerals correspond uniquely to components, are correctly described in the specification, and no extraneous or conflicting references are present. Violations include missing numerals, duplicated labels, mismatched descriptions, and references to non-existent figures. This structured procedure enables systematic detection of figure–reference inconsistencies in multi-figure patent documents.

² Preliminary qualitative comparisons among GPT-4o, DeepSeek, and Claude-4-Sonnet indicated more stable performance of Claude-4-Sonnet in handling long-form, structured patent documents; therefore, it was selected for the reported experiments.

3.4. Comprehensive evaluation & improvement guidance

The integration module consolidates outputs from the three detection modules into a document-level quality assessment.

Each detected issue is encoded as a structured record including its source module, risk severity, affected section, and supporting evidence. Based on these records, the module derives an overall quality profile reflecting the distribution and severity of detected defects. The module produces two outputs: (i) an overall filing-readiness classification — *Filing Ready*, *Minor Revision Needed*, *Major Revision Needed*, or *Significant Rework Required* — and (ii) a ranked list of revision actions aligned with defect severity and examination impact. Each action specifies the defect type, affected section, and revision objective. Representative integration outputs for both AI-generated and human-authored specifications are provided in [Appendix](#).

4. Experimental setup

4.1. Research design overview

We evaluate the proposed framework against expert-annotated ground truth on a dataset of 160 patent specifications (80 human-authored and 80 AI-generated) spanning eight IPC domains. All documents were anonymized during annotation and evaluation to ensure objective assessment.

The experimental design addresses the following research questions:

RQ1: What is the detection accuracy of our framework compared to expert annotations?

RQ2: How do defect patterns vary across patent sections and technical domains?

RQ3: What are the key quality differences between human-authored and AI-generated patents?

4.2. Validation protocol and ground truth

To establish reliable ground truth for evaluation, each patent document was independently annotated by two patent professionals with experience in CNIPA patent drafting and examination. Annotations were performed at the granularity aligned with our three modules: (i) regulatory compliance and technical coherence were labeled at the sentence level, and (ii) figure-reference consistency was labeled at the level of *figure-reference pairs*. Disagreements were resolved through joint review. When uncertainty persisted, conservative labels were adopted to minimize the risk of overlooking examination-critical defects.

4.3. Dataset construction

4.3.1. Sample selection criteria

Human-authored patents were curated as baseline references according to three criteria: (i) publication between 2015 and October 2022 to ensure temporal separation from recent large language model deployment, (ii) complete specifications with clear technical disclosure and comprehensive claim sets, and (iii) balanced coverage across eight IPC domains and two patent types (invention and utility model).

Eighty AI-generated patent specifications were produced using two Chinese commercial drafting tools, Eureka and MindFlowing. Each AI-generated document was paired with a corresponding human-authored patent to preserve balanced distribution across IPC domains and patent types ([Table 2](#)). For each pair, identical input materials — including abstracts, claims, technical descriptions, and figures — were provided to both tools. Eureka generates complete specifications based on all inputs, whereas MindFlowing produces textual specifications without integrated figures.

Table 2

Human patent distribution by IPC category and patent type.

IPC category	Invention	Utility model	Total
A - Human Necessities	5	5	10
B - Operations, Transport	5	5	10
C - Chemistry, Metallurgy	5	5	10
D - Textiles, Paper	5	5	10
E - Fixed Constructions	5	5	10
F - Mechanical Engineering	5	5	10
G - Physics	5	5	10
H - Electricity	5	5	10
Total	40	40	80

Table 3

Descriptive statistics by document source.

Metric	Human patents	Eureka	MindFlowing
Document length	5329.88 ± 4137.34	4577.35 ± 2907.45	2756.38 ± 586.12
Claims count	7.88 ± 3.22	7.47 ± 3.76	8.28 ± 2.56
Figure references	3.46 ± 2.39	3.20 ± 2.81	3.73 ± 1.88
Sample size (N)	80	40	40

4.3.2. Dataset characteristics

[Table 3](#) summarizes key descriptive statistics of the dataset. Human-authored patents exhibit substantial variability in document length (5330 ± 4137 words), while Eureka produces documents of comparable length (4577 ± 2907 words). In contrast, MindFlowing generates consistently shorter specifications with limited variation (2756 ± 586 words), indicating reduced adaptability to input complexity. Claims counts and figure references are broadly comparable across sources, supporting the paired comparison design.

4.4. Evaluation metrics

Given the pronounced class imbalance across all modules, including uneven compliance ratios and skewed risk-level distributions, we evaluate the three detection modules using balanced accuracy, precision, recall, and F1-score. Balanced accuracy is defined as the average of sensitivity and specificity and is adopted to mitigate the influence of dominant majority classes.

For the technical coherence module, which involves multi-class risk classification, we additionally report macro-averaged F1-scores to ensure equal weighting across risk categories:

$$\text{Macro-F1} = \frac{1}{C} \sum_{i=1}^C \frac{2P_i R_i}{P_i + R_i} \quad (1)$$

where C denotes the number of risk classes, and P_i and R_i represent the precision and recall for class i , respectively.

5. Results and analysis

5.1. Overall framework performance (RQ1)

To answer RQ1, we evaluate each module against expert annotations. [Table 4](#) reports balanced accuracy and class-wise precision/recall/F1.

5.1.1. Regulatory compliance detection performance

The regulatory compliance module was evaluated on 10,841 sentences (95.0% compliant; 5.0% non-compliant). It achieves $99.74 \pm 0.07\%$ balanced accuracy and perfect recall for non-compliance, indicating no violations are missed. Precision for non-compliant predictions is $91.17 \pm 2.11\%$, implying a manageable false-positive load for manual review. Compliant predictions remain highly reliable, with precision of $100.0 \pm 0.00\%$ and recall of $99.49 \pm 0.14\%$.

Error Analysis: False positives mainly come from borderline but formally acceptable phrasing (50%–55%), conservative handling of

Table 4
Framework detection performance.

Task & Category	N	Acc. (%)	Pre. (%)	Rec. (%)	F1 (%)
Regulatory Comp.	10,841	99.74 ± 0.07			
Compliant	10,298		100.0 ± 0.00	99.49 ± 0.14	99.74 ± 0.07
Non-compliant	543		91.17 ± 2.11	100.0 ± 0.00	95.19 ± 1.20
Technical coherence	8924	82.12 ± 2.28			
High risk	175		76.74 ± 5.49	81.92 ± 5.40	79.30 ± 4.15
Medium risk	297		68.61 ± 3.84	92.20 ± 3.31	78.98 ± 2.87
Low risk	206		89.97 ± 5.34	54.93 ± 6.83	68.36 ± 5.57
Safe	8246		99.95 ± 0.04	99.56 ± 0.13	99.74 ± 0.07
Figure consistency	554	91.2 ± 2.15			
Consistent	371		97.6 ± 1.80	86.8 ± 3.65	91.9 ± 2.70
Inconsistent	183		78.1 ± 5.85	95.6 ± 3.20	86.0 ± 4.35

Values shown as mean ±95% confidence interval half-width from bootstrap resampling (1000 iterations).

Table 5
Confusion matrix analysis for risk category misclassifications.

True\Predicted	High	Medium	Low	Safe
High risk (175)	144 (82.3%)	24 (13.7%)	7 (4.0%)	0 (0.0%)
Medium risk (297)	21 (7.1%)	274 (92.3%)	2 (0.7%)	0 (0.0%)
Low risk (206)	13 (6.3%)	76 (36.9%)	113 (54.9%)	4 (1.9%)
Safe (8246)	10 (0.1%)	23 (0.3%)	4 (0.0%)	8209 (99.6%)

Values in parentheses represent the proportion of true labels classified into each predicted category. Diagonal values (in bold) represent correct classifications.

nested/conditional structures (around 35%), and domain-specific terminology deviations (around 15%). These cases are predominantly low-severity and reflect a recall-oriented screening strategy rather than systematic misidentification of true violations.

Practical Implications: The conservative bias prioritizes recall over precision, aligning with patent prosecution practice where missed violations carry higher cost. The module is therefore suitable as a first-pass screening tool with targeted human verification.

5.1.2. Technical coherence validation performance

We evaluate technical coherence on 8924 technical sentences after removing non-technical templates. Table 5 reports the confusion matrix, which we use to analyze boundary errors between risk levels.

Despite substantial class imbalance, the module achieved a balanced accuracy of $82.12 \pm 2.28\%$, validating its practical applicability to real patent prosecution scenarios where safe content predominates. Meanwhile, the inherent complexity of technical risk assessment leads to significant performance variations across categories. Safe content is identified with near-perfect accuracy, enabling reliable automation of non-problematic sentences. Among risk categories, medium-risk content shows the strongest recall, while high-risk content achieves balanced precision and recall.

Error Analysis: As shown in Table 5, 17.7% of High-risk cases are under-escalated, primarily to Medium risk, while 45.1% of Low-risk cases are over-escalated, again mainly to Medium. Misclassifications concentrate at the Low-Medium boundary, reflecting a conservative bias rather than systematic failure to identify severe defects.

Practical Implications: High-confidence Safe predictions enable triage, while High-risk cases benefit from expert review due to remaining under-escalation.

5.1.3. Figure-reference consistency verification performance

The figure-reference consistency module was evaluated on 554 figure-reference pairs from 144 patents (371 consistent; 183 inconsistent). It achieves $91.2 \pm 2.15\%$ balanced accuracy. For inconsistent pairs, recall is high ($95.6 \pm 3.20\%$) with lower precision ($78.1 \pm 5.85\%$), indicating conservative flagging.

Error Analysis: Most false positives arise from structurally complex figures. Approximately half occur in dense numbering scenarios (>15

elements), while about 30% involve non-sequential or nested numbering patterns that hinder automated matching. The remaining errors are primarily attributable to figure quality issues, such as low resolution or overlapping callouts that obscure reference identification.

Practical Implications: The high recall for inconsistency detection (95.6%) ensures comprehensive identification of genuine figure-reference mismatches, critical for patent quality assurance. While the 21.9% false positive rate requires additional manual review, this conservative approach prevents overlooking actual inconsistencies. The 91.2% balanced accuracy supports automated deployment with human verification for complex reference scenarios exceeding 15 elements or involving non-standard numbering patterns.

5.2. Defect distribution analysis (RQ2)

We analyze defect rate variations across patent sections and IPC classes to identify structural and domain-specific factors influencing patent quality.

5.2.1. Patent section analysis

Fig. 2 shows that patent quality challenges manifest as distinct error patterns specific to each section's functional requirements.

The **Technical Field** section shows the lowest overall defect rate but concentrates high-risk errors, primarily patent type misclassification. In contrast, the **Background** and **Summary** sections are more prone to self-novelty negation, weak prior-art relevance, and insufficient technical detail. **Brief Description of Drawings** exhibits the highest defect rate at 23.56%, largely due to the concentration of figure-text consistency checks in this section for evaluation consistency. This high error rate highlights the critical need for enhanced multimodal capabilities in patent drafting tools. Apart from similar defect patterns in Summary, **Detailed Description** defects also include identical embodiment duplications and misuse of relational terms (e.g., “optional”, “further”). These errors significantly compromise patent validity by failing to demonstrate adequate technical variation and proper hierarchical relationships between embodiments.

Cross-sectional analysis reveals that maintaining terminological consistency and figure-text alignment across multiple sections represent fundamental challenges requiring priority attention.

5.2.2. Technical domain analysis

Fig. 3 reveals significant performance disparities across IPC domains. **Physics (Class G)** achieves superior performance with the lowest overall defect rate (2.12%), leading across non-compliance (2.14%), medium-risk (2.21%), and low-risk (1.35%) categories. This exceptional performance may be attributed to mathematical precision and standardized measurement protocols. Additionally, Physics patents typically focus on method-based inventions rather than complex apparatus descriptions, reducing reference numeral management complexity. **Mechanical Engineering (Class F)** and **Construction**

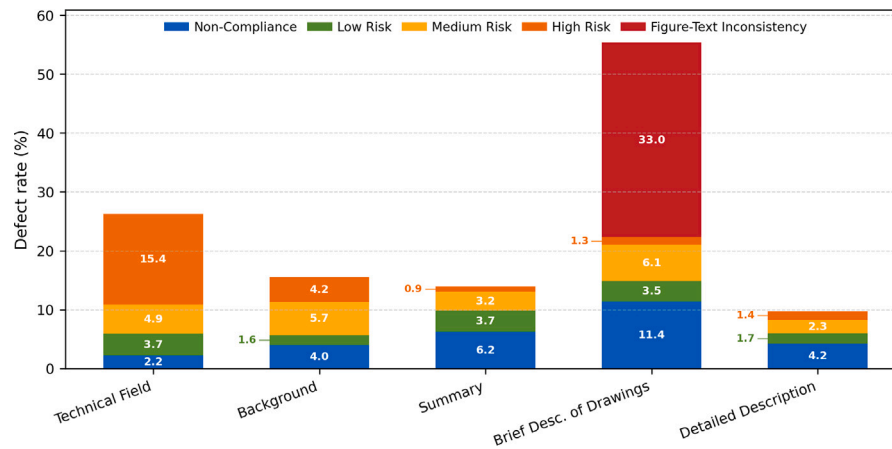


Fig. 2. Distribution of detected defects across patent specification sections. The stacked bars summarize regulatory non-compliance, technical risk levels, and figure-reference inconsistencies by section, highlighting structurally distinct quality risk profiles that motivate section-aware prioritization in the integration module.

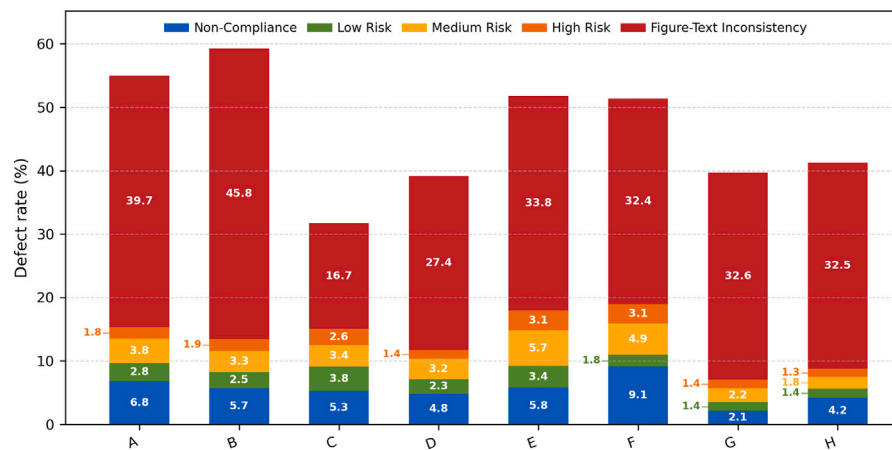


Fig. 3. Distribution of detected defects across IPC technical domains. The stacked bars summarize regulatory non-compliance, technical risk levels, and figure-reference inconsistencies by IPC classification, highlighting domain-specific quality risk profiles that motivate domain-aware analysis in the integration module.

(Class E) show elevated defect rates (5.45% and 5.09% respectively) due to their comprehensive documentation requirements. Mechanical Engineering demands detailed three-dimensional system descriptions with intricate component relationships, while Construction involves multi-disciplinary integration across structural, mechanical, electrical, and environmental subsystems. These requirements make claim-specification alignment and figure-text consistency particularly challenging. **Figure inconsistency** represents the most pervasive quality challenge, with rates ranging from 16.67% (Class C) to 45.83% (Class B) across all IPC classes, highlighting the critical need for enhanced multimodal patent generation capabilities. **Chemistry (Class C)** demonstrates moderate performance (3.96% defects) with the lowest figure inconsistency rate (16.67%), suggesting that chemical formulas and reaction schemes provide clearer structural guidance than mechanical drawings. **Electricity (Class H)** shows strong performance (2.77% defects), likely benefiting from standardized circuit symbols and well-established electrical engineering conventions.

Cross-domain analysis demonstrates that patent quality correlates with domain-specific factors: mathematical rigor (Physics), standardized conventions (Electricity), and documentation complexity (Mechanical Engineering, Construction). Figure-reference inconsistency consistently appears as a dominant quality challenge across IPC categories.

5.3. Comparative quality analysis (RQ3)

To answer RQ3, we compare human-authored and AI-generated specifications using average defects per document, which reflects the

revision burden experienced in practice while remaining comparable across variable-length documents. Table 6 reports pairwise differences with significance tests.

The human-AI gaps show statistical significance (H-E: -1.911 , H-M: -5.009 , $p < 0.05$), indicating patent drafting tools require ongoing enhancement. Figure-reference inconsistency represents the most pronounced quality gap across all comparisons (H-E: -1.073 , H-M: -3.271 , $p < 0.05$). The substantial performance difference between AI tools (E-M: -2.198 , $p < 0.05$) underscores the importance of multimodal capabilities in specification generation tasks. While Mind-Flowing's figure-free approach leads to systematic reference misalignment, Eureka's integrated figure generation achieves better coordination, though both require significant enhancement to reach human-level accuracy. Human-authored patents show notably more low-risk defects, primarily typographical mistakes and minor formatting issues that do not compromise technical understanding. Critically, both patent drafting tools exhibit more defects in higher-risk categories (H: 0.325, E: 1.900, M: 1.825). These defects primarily manifest as inconsistencies between patent sections and misalignment with claims, reflecting limitations in current system architectures. All sources show substantial regulatory compliance challenges, though the defect patterns differ significantly. Human-authored patents primarily contain typographical errors, punctuation mistakes, and minor commercial language issues that escaped editorial review. In contrast, AI-generated compliance issues predominantly involve inappropriate commercial language usage, claims of beneficial effects without adequate technical support, and

Table 6
AI tools vs. Human performance in patent drafting quality.

Dimension	H	E	M	H-E	H-M	E-M
Regulatory Comp.						
Non-compliance	2.700	4.400	3.775	-1.700 ^a	-1.075 ^a	+0.625 ^a
Technical coherence						
High risk	0.325	1.900	1.825	-1.575 ^a	-1.500 ^a	+0.075 ^a
Medium risk	2.050	1.050	2.275	+1.000 ^a	-0.225 ^a	-1.225 ^a
Low risk	1.913	0.475	0.850	+1.438 ^a	+1.062	-0.375 ^a
Figure-Text Cons.						
Figure-Ref Incons.	0.139	1.212	3.410	-1.073 ^b	-3.271 ^b	-2.198 ^b
Total defects per Doc	7.126	9.037	12.135	-1.911^b	-5.009^b	-3.098^b

Notes:
^a $p < 0.001$.
^b $p < 0.05$.

inconsistent patent classification terminologies that create regulatory ambiguity.

Overall, AI tools reduce drafting effort but still lag in high-risk technical consistency and figure-text integration. The observed ordering (Human > Eureka > MindFlowing) suggests that tighter multimodal coupling and stronger global consistency constraints are key to closing the quality gap.

6. Conclusion

This paper presents a multi-dimensional LLM-based framework for automated quality assurance of patent specifications. The framework addresses scalability and consistency limitations of expert-driven examination in the context of AI-assisted patent drafting. Unlike prior studies that focus on isolated components such as claims, our approach evaluates patent specifications as complete documents and assesses quality along three dimensions: regulatory compliance, technical coherence, and figure-text consistency.

Experiments on 160 patent specifications, including both human-authored and AI-generated documents, demonstrate robust agreement with expert-annotated ground truth across all modules. Comparative analysis reveals systematic quality disparities between human and AI drafting. Although current AI tools improve drafting efficiency, they exhibit disproportionately higher defect rates in high-risk technical consistency and figure-text alignment. Among all defect categories, figure-text inconsistency emerges as the most persistent bottleneck, highlighting the importance of multimodal integration in patent specification generation and evaluation.

Several limitations remain. First, the empirical evaluation is based on CNIPA specifications, and extension to other jurisdictions requires incorporation of jurisdiction-specific examination criteria. Second, fine-grained risk discrimination, particularly near the low-medium boundary, warrants further refinement. Third, dense and irregular figure-reference structures continue to challenge automated multimodal consistency verification. Addressing these issues and expanding the framework to broader drafting environments constitute important directions for future work.

Overall, this study establishes a structured foundation for document-level patent quality assessment and provides empirical evidence for understanding the strengths and limitations of AI-assisted patent drafting.

CRediT authorship contribution statement

Yuqian Chai: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Investigation, Formal analysis, Data curation. **Chaochao Wang:** Writing – review & editing, Project administration, Conceptualization. **Weilei Wang:** Writing – review & editing, Project administration.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude-4-Sonnet to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

All authors are employed by PatSnap, one of the companies whose AI patent drafting tools was evaluated in this study. However, none of the authors are affiliated with PatSnap’s patent drafting tool development team or were involved in the development of the evaluated features.

The authors’ employment relationship with PatSnap did not influence the study design, methodology, data collection, analysis, or conclusions. This research was conducted independently with the objective of providing an unbiased comparative analysis of available AI patent drafting tools in the market. The study methodology was designed to ensure objectivity, and the evaluation criteria were applied consistently across all platforms examined.

The authors declare no other competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. Illustrative integration outputs

This appendix presents representative document-level outputs produced by the integration module. The examples are anonymized and translated for academic reporting purposes.

Example A: AI-generated specification

Overall assessment
Filing Readiness: Significant Rework Required

- **Total Segments:** 98
- **Compliance Rate:** 90.8% (89/98 segments compliant)
- **Risk Distribution:** High-risk: 3 | Medium-risk: 2 | Low-risk: 4 | Safe: 89

High-risk issues
Issue 1: Self-defeating novelty disclosure.

- **Problem:** Cited patent CN2437116Y describes a solution identical to Claim 1, directly undermining patentability.
- **Recommendation:** Re-verify the citation; if accurate, redesign claims with distinguishing features.

Issue 2: Verbatim section duplication.

- **Problem:** Summary section copies Claim 1 verbatim—a typical AI-generation artifact.
- **Recommendation:** Rewrite Summary to provide technical context rather than claim replication.

Issue 3: Fabricated technical effects.

- **Problem:** Claims of “significantly improved efficiency” lack supporting data or analysis.
- **Recommendation:** Provide quantitative evidence or remove unsupported assertions.

Medium-risk issues
Issue 1: Figure-text mismatch.

- **Problem:** Corrupted character sequences in figure descriptions indicate encoding errors or AI artifacts.
- **Recommendation:** Correct all corrupted text and verify figure-text correspondence.

Issue 2: Inconsistent reference numerals.

- **Problem:** Conflicting numbering (e.g., pedal and frame both labeled “5”).
- **Recommendation:** Create comprehensive numbering scheme and ensure consistency.

Low-risk issues

Issue 1: Subjective language. Use of phrases like “simple structure, excellent effect” violates objective description requirements.

Issue 2: Exaggerated claims. Phrases like “significantly reduced labor intensity” lack qualification or comparative context.

Issue 3: Formulaic descriptions. Repetitive sentence structures typical of AI-generated text reduce naturalness.

Issue 4: Incomplete prior art analysis. Background lacks detailed examination of existing solutions and their specific limitations.

Revision recommendations

Priority 1 (Critical):

1. Resolve novelty disclosure issue—verify CN2437116Y and redesign claims if necessary.
2. Rewrite Summary section to eliminate verbatim claim duplication.
3. Remove or substantiate all unsupported technical effect claims.

Priority 2 (Important):

1. Correct corrupted text and resolve reference numeral conflicts.
2. Remove subjective and exaggerated language throughout.

Priority 3 (Recommended):

1. Expand prior art analysis with specific technical comparisons.
2. Add technical details to reduce formulaic descriptions.

Conclusion

This specification contains critical defects affecting patentability, particularly the self-defeating novelty disclosure. Multiple AI-generation indicators (verbatim duplication, fabricated citations, formulaic descriptions) suggest substantial human expert intervention is required. **Do not file** until all Priority 1 issues are resolved and comprehensive expert review is completed.

Example B: Human-authored specification

Overall assessment

Filing Readiness: Minor Revision Recommended

- **Total Segments:** 18
- **Compliance Rate:** 94.4% (17/18 segments compliant)
- **Risk Distribution:** High-risk: 0 | Medium-risk: 1 | Low-risk: 3 | Safe: 13

High-risk issues

No high-risk issues detected. The document demonstrates strong technical coherence, appropriate claim-specification correspondence, and no indicators of fabricated content or critical legal defects.

Medium-risk issues

Issue 1: Insufficient prior art analysis.

- **Problem:** Background addresses only surface-level problems without analyzing specific existing solutions or their fundamental limitations, weakening the technical argumentation for non-obviousness.
- **Recommendation:** Expand to include detailed description of existing tools, analysis of why they fail, and technical explanation of root causes that the present invention addresses.

Low-risk issues

Issue 1: Minor language error. Duplicated word in Summary section describing labor intensity reduction.

Issue 2: Terminology inconsistency. Minor variation between specification (“preferably provided with umbrella frame”) and claims (“provided with umbrella frame”).

Issue 3: Ambiguous technical term. Potential confusion between “flower disk” and “chain disk” used in different sections—unclear if these refer to the same or different components.

Revision recommendations

Priority 1 (Important):

1. Expand background with detailed prior art analysis to strengthen non-obviousness argument.
2. Correct duplicated word and unify terminology between specification and claims.

Priority 2 (Recommended):

1. Verify and standardize technical terms (“flower disk” vs. “chain disk”).

Conclusion

This specification demonstrates good overall quality with complete technical solutions, clear logical structure, and strong claim-specification correspondence (>95%). Minimal AI-generation risk is evident from detailed implementation descriptions, natural technical flow, and varied sentence structures. Primary issues concern language precision and technical argumentation depth rather than fundamental defects. **Ready for filing** after addressing Priority 2 items; good prospects for allowance expected.

Data availability

Data will be made available on request.

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Dr. Yuqian Chai, a Senior Natural Language Processing (NLP) Engineer at Patsnap Information Technology Co., Ltd. She holds a Ph.D. in Electrical and Electronic Engineering from The University of Hong Kong. Her research interests include artificial intelligence applications in intellectual property, patent analytics, AIGC detection and patent drafting.

Mr. Chaochao Wang, a Computer Technology professional holding a Master's degree, currently serves as the Natural Language Processing (NLP) Manager at Patsnap Information Technology Co., Ltd. With over ten years of dedicated research experience in AI algorithms within the Intellectual Property domain, Mr. Wang specializes in the application of artificial intelligence to IP solutions. His expertise extends to spearheading the development of key AI applications, including Semantic Search, Machine Translation, Novelty Search, and AIGC detection, reflecting his deep commitment to technical innovation in the IP industry.

Mr. Weilei Wang currently serves as the Chief AI Scientist at PatSnap, holding a Master's degree in Computer Technology. With over 15 years of dedicated research experience in AI algorithms within the Intellectual Property domain, he specializes in the development of AI-driven IP applications. His expertise extends to spearheading initiatives in Information Extraction, Patent Drafting, and AIGC detection technologies.